since the rewards an individual agent receives will vary not only based on its own actions but on the actions of other agents. This is because all agents are reinforcement learning agents that learn through experience; their policies are constantly changing in response to changes in the environment. If we use IL-Q in this nonstationary environment, we lose the convergence guarantee, and this can impair the performance of independent Q-learning significantly.

**Figure 9.3. In a stationary environment, the expected (i.e., average) value over time for a given state will remain constant (stationary). Any particular state transition may have a stochastic component, hence the noisy-looking time series, but the mean of the time series is constant. In a nonstationary environment, the expected value for a given state transition will change over time, which is depicted in this time series as a changing mean or baseline over time. The Q function is trying to learn the expected value for state-actions, and it can only converge if the state-action values are stationary, but in the multi-agent setting, the expected state-action values can change over time due to the evolving policies of other agents.**